Imperial College London

Department of Electronic and Electrical Engineering

Examinations 2023

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

May, 2023 • Exam Duration: Three Hours

There are FOUR questions on this paper.
Answer Question 1 and any TWO other questions.
Question 1 is worth 40% of the marks and other questions are worth 30%.

Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Tania Stathaki.

Question 1 [40 Marks]

a) Real-world signals are **[2 Marks]**

(a) analog in amplitude,

(b) quantized in amplitude,

(c) vary continuously in time,

(d) are discrete in time.

Real-world signals are analog in amplitude and vary continuously in time.

b) How is a discrete signal scaled so that the sampling frequency satisfies $f_s = 1$? If the original analog signal has bandwidth Ω rad/s the scaled samples represent samples from an analog signal of what bandwidth? **[2 Marks]**

To achieve the desired scaling all frequencies are divided by $f_s = 1/T$, where T is the sampling period. This has the effect of dividing time samples by T. In the analog domain, the effect is of dividing the bandwidth by $1/T$ yielding the bandwidth $T\Omega$. When sampling at Nyquist rate, this results in bandwidth π rad/s.

c) Prove that discrete convolution is associtive and commutative. **[2 Marks]**

We have that

$$x[n] * (v[n] * w[n]) = \sum_p x[p] \sum_k v[k] w[(n-p)-k] = \sum_k v[k] \sum_p x[p] w[(n-k)-p]$$
$$= v[n] * (x[n] * w[n]).$$

Furthermore,

$$v[n] * w[n] = \sum_k v[k] w[n-k] = \sum_p v[n-p] w[p]. \tag{1}$$

In the last equality we computed the change of variable $p = n - k$.

d) Let $H(e^{\jmath\omega})$ be a zero-phase filter of odd length $M+1$, symmetric around 0, i.e., $h[-n] = h[n]$. The weighted error is $e(\omega) = s(\omega)\,(H(e^{\jmath\omega} - d(\omega))$, where $d(\omega)$ is the target filter satisfying

$$d(\omega) = \begin{cases} 1, & 0 \le \omega \le \omega_1, \\ 0, & \omega_2 \le \omega \le \pi. \end{cases} \tag{2}$$

Choose $s(\omega)$ to control the error variation with ω. **[2 Marks]**

Filter $d(\omega)$ is a low-pass filter. A potential realisation for $s(\omega)$ is

$$s(\omega) = \begin{cases} \delta^{-1}, & 0 \le \omega \le \omega_1, \\ \epsilon^{-1}, & \omega_2 \le \omega \le \pi. \end{cases} \tag{3}$$

e) Let system $G(z)$ be defined as

$$G(z) = \frac{8 - 2z^{-1}}{4 - (2+\alpha)z^{-1} + 3z^{-2}}. \tag{4}$$

Find $\alpha \in \mathbb{R}$ such that $G(z)$ has two real poles. **[1 Marks]**

The poles of $G(z)$ are the solution of $4z^2 - (2+\alpha)z + 3 = 0$. The discriminant satisfies $\Delta = \alpha^2 + 4\alpha - 44$ which should be positive to ensure the poles are real. For this, we compute the roots of the discriminant and conclude that it is positive when α is not between the roots such that

$$\alpha \in \left(-\infty, \frac{-4-\sqrt{192}}{2}\right) \cup \left(\frac{-4+\sqrt{192}}{2}, \infty\right). \tag{5}$$

f) The input/output relation for a signal processing system is described by,

$$2u\,[n] \longrightarrow \boxed{h\,[n]} \longrightarrow \left(\frac{2^2}{2^n} - 3\left(-\frac{3}{4}\right)^n\right) u\,[n]$$

where $u[n]$ is the usual unit-step sequence.

Find the impulse response of this system. **[3 Marks]**

Let us write the system as,

$$\underbrace{2u\,[n]}_{x[n]} \longrightarrow \boxed{h\,[n]} \longrightarrow \underbrace{\left(4\frac{1}{2^n} - 3\left(-\frac{3}{4}\right)^n\right) u\,[n]}_{y[n]}.$$

Now, observe that,

$$\delta\,[n] = u\,[n] - u\,[n-1] = \frac{x\,[n] - x\,[n-1]}{2}.$$

Hence, we can evaluate the impulse response of the system by feeding in,

$$\delta\,[n] = \frac{x\,[n] - x\,[n-1]}{2} \longrightarrow \boxed{h\,[n]} \longrightarrow \frac{y\,[n] - y\,[n-1]}{2}$$

which gives,

$$\frac{y\,[n] - y\,[n-1]}{2} = \frac{1}{2}\left(4\frac{1}{2^n} - 3\left(-\frac{3}{4}\right)^n\right) u\,[n] - \frac{1}{2}\left(4\frac{1}{2^{n-1}} - 3\left(-\frac{3}{4}\right)^{n-1}\right) u\,[n-1]$$

or,

$$h\,[n] = \frac{1}{2}\delta\,[n] + \left(-2\left(\frac{1}{2}\right)^n - \frac{7}{2}\left(-\frac{3}{4}\right)^n\right) u\,[n-1].$$

g) Suppose that $s\,(t)$ is a bandlimited signal with maximum frequency of $\Omega = \pi$ (rad/sec).

(a) What is the largest sampling period T that allows for reconstruction of $s(t)$ from $s[k] = s(kT)$? **[1 Marks]**

From Nyquist-Shannon Theorem, $T \leq \pi/\Omega$. Hence $T \leq 1$ suffices reconstruction.

(b) Assume that we are given samples $s[k]$, derive an algorithm to obtain τ-delayed samples,

$$s_\tau[k] = s(kT - \tau)$$

where $0 < \tau < T$. **[3 Marks]**

Since s is bandlimited, according to the Nyquist-Shannon formula, we have,

$$s(t) = \sum_{n\in\mathbb{Z}} s[n]\,\mathrm{sinc}(t-n), \quad T = 1$$

and hence,

$$s(t-\tau) = \sum_{n\in\mathbb{Z}} s[n]\,\mathrm{sinc}(t-k-\tau).$$

Re-sampling the above equation at $t = kT, T = 1$,

$$s(t-\tau)|_{t=kT,T=1} = \sum_{n\in\mathbb{Z}} s[n]\,\mathrm{sinc}(k-n-\tau) \Rightarrow \boxed{s_\tau[k] = s[k] * \mathrm{sinc}(k-\tau)}.$$

h) A filter is defined by its complex-valued pole $\lambda_0 \in \mathbb{C}$,

$$\frac{B(z,\lambda_0)}{A(z,\lambda_0)} = \frac{z^{-1} - \lambda_0^*}{1 - \lambda_0 z^{-1}}$$

where λ_0^* is the complex-conjugate of λ_0.

(a) Show that $B(z,\lambda_0)/A(z,\lambda_0)$ defines an all-pass filter. **[1 Marks]**

To show that this system is an all-pass filter, note that the frequency response is

$$H(e^{\jmath\omega}) = H(z)|_{z=e^{\jmath\omega}} = \frac{e^{-\jmath\omega} - a^*}{1 - ae^{-\jmath\omega}}$$

Therefore, the squared magnitude is

$$|H(e^{\jmath\omega})|^2 = H(e^{\jmath\omega})H^*(e^{\jmath\omega}) = \frac{e^{-\jmath\omega} - a^*}{1 - ae^{-\jmath\omega}} \cdot \frac{e^{\jmath\omega} - a}{1 - a^* e^{\jmath\omega}} = \frac{1 + |a|^2 - 2\Re(a^* e^{\jmath\omega})}{1 + |a|^2 - 2\Re(a^* e^{\jmath\omega})} = 1$$

and $H(e^{\jmath\omega})$ is an all-pass filter.

(b) Show that the cascaded representation,

$$\text{Input} \longrightarrow \underbrace{\boxed{\frac{B(z,\lambda_0)}{A(z,\lambda_0)}} \longrightarrow \boxed{\frac{B(z,\lambda_0^*)}{A(z,\lambda_0^*)}}}_{H(z)} \longrightarrow \text{Output}$$

allows for an implementation using real-valued coefficients only. Letting,

$$H(z) = \frac{\sum_{m=0}^{M-1} b_m z^{-m}}{\sum_{m=0}^{M-1} a_m z^{-m}}$$

identify M, a_m and b_m. **[3 Marks]**

Clearly,

$$\begin{aligned} H(z) &= \frac{B(z,\lambda_0)}{A(z,\lambda_0)}\frac{B(z,\lambda_0^*)}{A(z,\lambda_0^*)} \\ &= \left(\frac{z^{-1}-\lambda_0^*}{1-\lambda_0 z^{-1}}\right)\left(\frac{z^{-1}-(\lambda_0^*)^*}{1-(\lambda_0)^* z^{-1}}\right) = \frac{z^{-2}-\lambda_0^* z^{-1}-\lambda_0 z^{-1}+\lambda_0^*\lambda_0}{1-\lambda_0^* z^{-1}-\lambda_0 z^{-1}+\lambda_0^*\lambda_0 z^{-2}} \\ &= \frac{z^{-2}-2z^{-1}\Re(\lambda_0)+|\lambda_0|^2}{|\lambda_0|^2 z^{-2}-2z^{-1}\Re(\lambda_0)+1} \end{aligned}$$

which is indeed a filter with only real-valued coefficients:
$b_m = \left\{|\lambda_0|^2, -2\Re(\lambda_0), 1\right\}$ in the numerator, and,
$a_m = \left\{1, -2\Re(\lambda_0), |\lambda_0|^2\right\}$ in the denominator.

(c) For input sequence $x[k]$, write the filtered output $y[k]$ in terms of the real-valued coefficients of filter $H(z)$ above.

[1 Marks]

In the z-transform domain, we have,

$$\frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^{M-1} b_m z^{-m}}{\sum_{m=0}^{M-1} a_m z^{-m}} \Rightarrow X(z)\sum_{m=0}^{M-1} b_m z^{-m} = Y(z)\sum_{m=0}^{M-1} a_m z^{-m}.$$

This simplifies to,

$$\begin{array}{c} X(z)|\lambda_0|^2 - 2\Re(\lambda_0)X(z)z^{-1} + X(z)z^{-2} = Y(z) - 2\Re(\lambda_0)Y(z)z^{-1} + |\lambda_0|^2 Y(z)z^{-2} \\ \Updownarrow \\ x[k]|\lambda_0|^2 - 2\Re(\lambda_0)x[k-1] + x[k-2] = y[k-2]|\lambda_0|^2 - 2\Re(\lambda_0)y[k-1] + y[k] \end{array}.$$

i) **Hilbert Relations.** Let us define a sequence,

$$s[k] = \begin{cases} 0 & k = 0 \\ \frac{k}{|k|} & k \neq 0 \end{cases}.$$

(a) By letting $S(z) = \sum_k s[k]z^{-k}$, show that $\left(1-z^{-1}\right)S(z) = 1+z^{-1}$. **[2 Marks]**

From the definition of $s[k]$, we have

$$s[k] = u[k-1] - u[-1-k]$$

and Hence, z-transform of $s[k]$ is given by

$$S(z) = \frac{z^{-1}}{1-z^{-1}} - \frac{z}{1-z} = \frac{1+z^{-1}}{1-z^{-1}}$$

which leads to the statement $(1-z^{-1})S(z) = 1+z^{-1}$.

(b) Evaluate $S\left(e^{\jmath\omega}\right)$. **[1 Marks]**

$$S(e^{\jmath\omega}) = \frac{1+e^{-\jmath\omega}}{1-e^{-\jmath\omega}} = \frac{e^{\jmath\frac{\omega}{2}}+e^{-\jmath\frac{\omega}{2}}}{e^{\jmath\frac{\omega}{2}}-e^{-\jmath\frac{\omega}{2}}} = \frac{2\cos(\frac{\omega}{2})}{2\jmath 2sin(\frac{\omega}{2})} = -\jmath\cot(\frac{\omega}{2}).$$

(c) Let $\overline{h}(t) = h(-t)$.

Based on this, let us define the following,

$$h_E[n] = \frac{1}{2}\left(h[n] + \overline{h}[n]\right) \quad \text{and} \quad h_O[n] = \frac{1}{2}\left(h[n] - \overline{h}[n]\right).$$

leading to Fourier relations, $\Re\left(H\left(e^{\jmath\omega}\right)\right) = H_E\left(e^{\jmath\omega}\right)$ and $\Im\left(H\left(e^{\jmath\omega}\right)\right) = -\jmath H_O\left(e^{\jmath\omega}\right)$. Show that whenever $h[n]$ is a causal sequence, it follows that real and imaginary parts of the spectrum of $h[n]$ are related to each other. In particular,

$$\Im\left(H\left(e^{\jmath\omega}\right)\right) = -\Re\left(H\left(e^{\jmath\omega}\right)\right) \odot \cot\left(\frac{\omega}{2}\right)$$

where $\odot$ denotes circular convolution. **[2 Marks]**

$$\Re(H(e^{\jmath\omega})) = H_e(e^{\jmath\omega}), \Im(H(e^{\jmath\omega})) = -\jmath H_o(e^{\jmath\omega})$$

Given that $h[k]$ is casual, we have

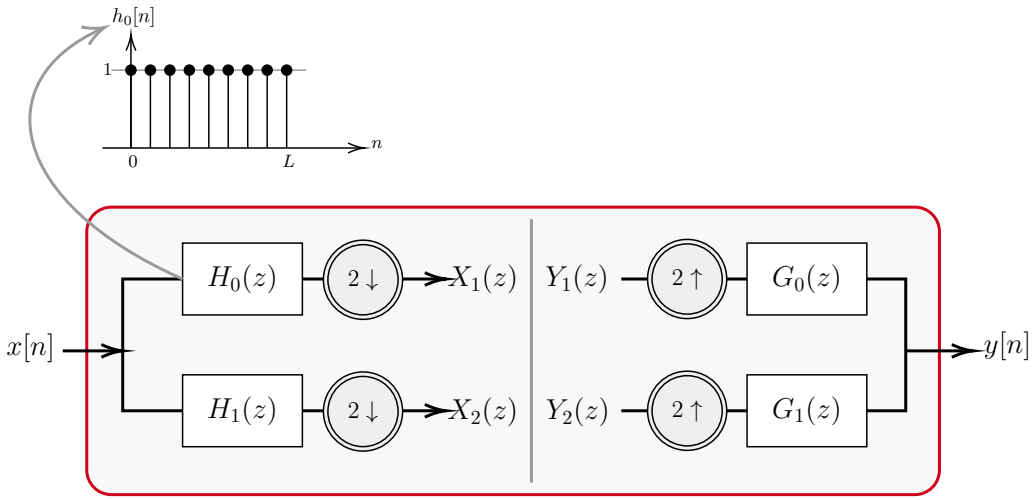
$$h_o[k] = h_e[k]s[k],\ h_e[k] = h[0]\delta[k] + h_o[k]s[k].$$

Hence, for causal signal $h[k]$:

$$\Im(H(e^{\jmath\omega})) = -\jmath(\Re(H(e^{\jmath\omega})) \circledast -\jmath\cot(\frac{\omega}{2})) = -\Re(H(e^{\jmath\omega})) \circledast \cot(\frac{\omega}{2}),$$
$$\Re(H(e^{\jmath\omega})) = H(\infty) + \jmath\Im(H(e^{\jmath\omega})) \circledast -\jmath\cot(\frac{\omega}{2}) = H(\infty) + \Im(H(e^{\jmath\omega})) \circledast \cot(\frac{\omega}{2})$$

This completes the proof.

j) The input-output relationship for a particular signal processing system is described by,



where the filter $H_0(z) = \sum_{n=0}^{L} z^{-n}$.

The input signal $x[n]$ undergoes filtering and decimation or downsampling (**2** $\downarrow$). This is the signal analysis part. There on, given sequences or transfer functions, the synthesis part reconstructs the original signal by upsampling (**2** $\uparrow$) followed by filtering. In the above H_0 and H_1 are low-pass and high-pass filters, respectively.

i) Identify outputs $X_1(z)$ and $X_2(z)$ given $x[n]$, $H_0(z)$ and $H_1(z)$. **[2 Marks]**

ii) How is $y[n]$ related to inputs transfer functions $Y_1(z)$ and $Y_2(z)$? **[2 Marks]**

iii) Suppose that one system's output becomes another system's input, that is,

$$Y_1(z) = X_1(z) \quad \text{and} \quad Y_2(z) = X_2(z).$$

What is the frequency response of $y[n]$? **[2 Marks]**

Under what conditions (in frequency domain), $y[n] = x[n]$? **[2 Marks]**

Let $h[n]$ be the impulse response of some filter $H(z)$. Furthermore, let $H_0(\omega) = H(\omega)$ and $H_1(\omega) = H(\omega - \pi)$. Identify the impulse response of $h_1[n]$, $g_0[n]$ and $g_1[n]$ in terms of $h[n]$. **[2 Marks]**

This is a classic textbook example also covered in details the notes and in the lecture, hence, short explanations are provided.

i) Here,

$$X_m(z) = \frac{1}{2}\left(V_m\left(z^{\frac{1}{2}}\right) + V_m\left(-z^{\frac{1}{2}}\right)\right), \qquad k = 0, 1$$

where $V_m(z) = H_m(z) X(z), m = 0, 1$.

ii)

$$Y(z) = Y_1\left(z^2\right) G_0(z) + Y_2\left(z^2\right) G_1(z).$$

iii) When $Y_1(z) = X_1(z)$ and $Y_2(z) = X_2(z)$, we have,

$$\begin{aligned} Y &= \frac{1}{2}\begin{bmatrix} X & \overline{X} \end{bmatrix}\begin{bmatrix} H_0 & H_1 \\ \overline{H}_0 & \overline{H}_1 \end{bmatrix}\begin{bmatrix} G_0 \\ G_1 \end{bmatrix} \\ &= \frac{1}{2}\left(X\left(H_0G_0 + H_1G_1\right) + \overline{X}\left(\overline{H}_0G_0 + \overline{H}_1G_1\right)\right) \end{aligned}$$

where $\overline{X}(z) = X(-z)$. Substituting $z = e^{\jmath\omega}$, we obtain the frequency response.

From the above part, we have a direct relation between $Y(z)$ and $X(z)$. For $y[n] = x[n]$ (perfect reconstruction), we can impose the follwing conditions,

$$Y = X\frac{1}{2}\underbrace{\left(H_0G_0 + H_1G_1\right)}_{\propto z^{-d}} + \overline{X}\frac{1}{2}\underbrace{\left(\overline{H}_0G_0 + \overline{H}_1G_1\right)}_{=0}$$

and hence,

$$\frac{1}{2}\left(H_0(z) G_0(z) + H_1(z) G_1(z)\right) = z^{-d} \quad \text{and} \quad \frac{1}{2}\left(\overline{H}_0(z) G_0(z) + \overline{H}_1(z) G_1(z)\right) = 0.$$

Again, replacing z by $e^{\jmath\omega}$ results in the desired conditions in the frequency domain.

In short, the filters are identified as,

$$\begin{bmatrix} h_0[n] & h_1[n] \\ g_0[n] & g_1[n] \end{bmatrix} = \begin{bmatrix} h[n] & (-1)^n h[n] \\ 2h[n] & -2(-1)^n h[n] \end{bmatrix}.$$

Detailed explanation is below.

We are given,

$$\begin{aligned} H_0(\omega) = H(\omega) &\Rightarrow H_0(z) = H(z) \\ H_1(\omega) = H(\omega - \pi) &\Rightarrow H_1(z) = H(-z) \end{aligned}.$$

From this, we obtain, $h_0[n] = h[n]$ and $h_1[n] = (-1)^n h[n]$, which is because,

$$H_1(z) = H(-z) = \sum\nolimits_n h_n(-z)^{-n} = \begin{cases} +h[n]z^{-n} & n \text{ is even} \\ -h[n]z^{-n} & n \text{ is odd} \end{cases}.$$

To compute the coefficients of filters g_0 and g_1, we make use of the equation,

$$Y = X\frac{1}{2}\underbrace{(H_0G_0 + H_1G_1)}_{\propto z^{-d}} + \overline{X}\frac{1}{2}\underbrace{(\overline{H}_0G_0 + \overline{H}_1G_1)}_{=0}$$

which can be re-written as,

$$\frac{1}{2}\begin{bmatrix} H_0 & H_1 \\ \overline{H}_0 & \overline{H}_1 \end{bmatrix}\begin{bmatrix} G_0 \\ G_1 \end{bmatrix} = \begin{bmatrix} z^{-d} \\ 0 \end{bmatrix} \xrightarrow{H_1=\overline{H}_0, H_0=H} \frac{1}{2}\begin{bmatrix} H & \overline{H} \\ \overline{H} & H \end{bmatrix}\begin{bmatrix} G_0 \\ G_1 \end{bmatrix} = \begin{bmatrix} z^{-d} \\ 0 \end{bmatrix}.$$

The above system of equations can be solved as simultaneous equations and this yields,

$$\begin{bmatrix} G_0 \\ G_1 \end{bmatrix} = \frac{2z^{-d}}{H\overline{H} - \overline{H}H}\begin{bmatrix} H & -\overline{H} \\ -\overline{H} & H \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} G_0 \\ G_1 \end{bmatrix} \propto \begin{bmatrix} +2H \\ -2\overline{H} \end{bmatrix}.$$

Hence $g_0[n] = 2h[n]$ and $g_1[n] = -2(-1)^n h[n]$.

k) Let

$$X[k] = \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi}{4N}(2n+1)k\right)$$

define the Discrete Cosine Transform of some sequence $x[n]$. Prove the anti-periodic and periodic property of $X[n]$, that is,

$$\underbrace{X[k+2N] + X[k] = 0}_{\text{Anti-Periodic}} \quad \text{and} \quad \underbrace{X[k+4N] - X[k] = 0}_{\text{Periodic}}.$$

[4 Marks]

For the first part (anti-periodic), we note that,

$$\cos\left(\frac{2\pi}{4N}(2n+1)k\right) \xrightarrow{k+2N} \frac{2\pi(2n+1)k}{4\pi} + 2\pi n + \pi$$

and since $\cos(\phi + \pi) = -\cos(\phi)$, we conclude that,

$$\cos\left(\frac{2\pi}{4N}(2n+1)k\right) \xrightarrow{k+2N} -\cos\left(\frac{2\pi}{4N}(2n+1)k\right)$$

and hence, $X[k+2N] = -X[k]$.

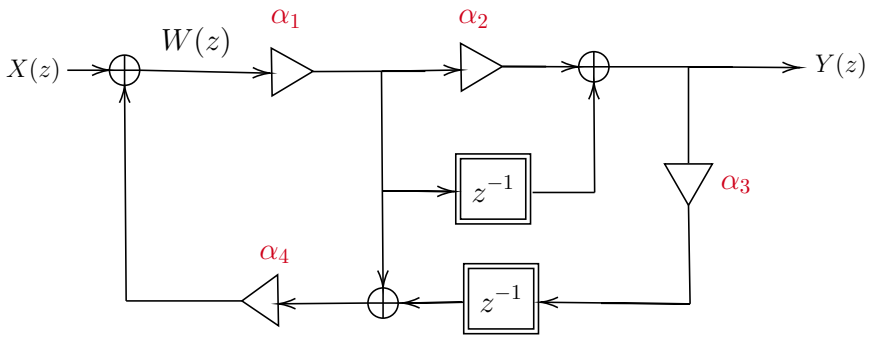
With the anti-periodic property, we have,

$$X[k+4N] = X[\underbrace{k+2N+2N}_{4N=2N+2N}] \xrightarrow{X[k+2N]=-X[k]} -X[k+2N] = X[k].$$

This shows that indeed $X[k+4N] - X[k] = 0$.

Question 2 [30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $W(z)$ in terms of $X(z)$. **[3 Marks]**

ii) Derive the expression of the transfer function $H(z) = \frac{Y(z)}{X(z)}$. **[4 Marks]**

iii) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[1 Mark]**

iv) Sketch the pole-zero plot for $\alpha_1 = \alpha_3 = 1, \alpha_2 = 2, \alpha_4 = 4$. **[1 Mark]**

i) We have that

$$\left[\alpha_3\left(W(z)\,\alpha_1\alpha_2 + W(z)\,\alpha_1 z^{-1}\right) + W(z)\,\alpha_1\right]\alpha_4 + X(z) = W(z)\,, \tag{6}$$

which results in

$$W(z) = \frac{1}{1 - \alpha_1\alpha_4 - \alpha_1\alpha_2\alpha_3\alpha_4 z^{-1} - \alpha_1\alpha_3\alpha_4 z^{-2}} X(z)\,. \tag{7}$$

ii) The output satisfies

$$\left(\alpha_1\alpha_2 + \alpha_1 z^{-1}\right) W(z) = Y(z) \tag{8}$$

Therefore the output satisfies

$$H(z) = \frac{\alpha_1\alpha_2 + \alpha_1 z^{-1}}{1 - \alpha_1\alpha_4 - \alpha_1\alpha_2\alpha_3\alpha_4 z^{-1} - \alpha_1\alpha_3\alpha_4 z^{-2}}. \tag{9}$$

iii) The poles are

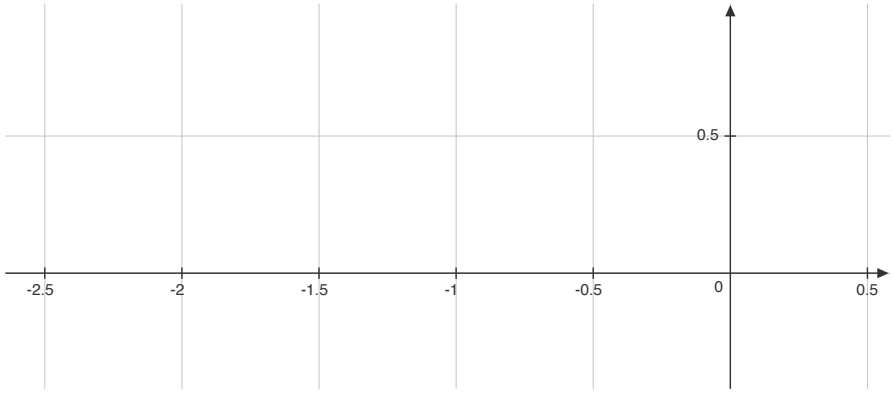
$$z_{1,2} = \frac{\alpha_1\alpha_2\alpha_3\alpha_4 \pm \sqrt{(\alpha_1\alpha_2\alpha_3\alpha_4)^2 + 4(1 - \alpha_1\alpha_4)\alpha_1\alpha_3\alpha_4}}{2(1 - \alpha_1\alpha_4)}. \tag{10}$$

Then the system is BIBO-stable if the poles are inside the unit circle $|z_{1,2}| < 1$.

iv) In this case

$$H(z) = \frac{2 + z^{-1}}{-3 - 8z^{-1} - 4z^{-2}}.$$

Here we have one zero $z_0 = -\frac{1}{2}$ and two pole $z_p^1 = -2, z_p^2 = -\frac{2}{3}$. The diagram is displayed below.



b) *i)* Let α is a complex-valued constant, compute the z-transform of a casual sequence $x[n] = e^{\alpha^2 n}$ and specify its region of convergence. **[2 Marks]**

ii) Suppose that $G_1(z) = 1 - \alpha z^{-1}$ and $G_2(z) = 1 - \beta z^{-1}$ where the constants α and β may be complex. If $\alpha\beta^* = 1$, show that $|G_1(e^{\jmath\omega})| = p|G_2(e^{\jmath\omega})|$ for all ω and determine the constant p. **[3 Marks]**

iii) Let $H_1(z) = 3z^{-2} - 16z^{-1} - 12$. Find the expression of $H_2(z)$ such that $|H_1(e^{\jmath\omega})| = |H_2(e^{\jmath\omega})|$ for all ω and that all the zeros of $H_2(z)$ lie inside the unit circle. **[3 Marks]**

iv) The impulse response, $h[n]$, of a linear time-invariant system satisfies $\sum_{n=-\infty}^{\infty} |h[n]| = S < \infty$. Show that the system is BIBO-stable and also that $H(z)$ converges for $|z| = 1$. **[3 Marks]**

i) From the problem description, we have (summing from $n = 0$ since $x[n]$ is causal)

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} = \sum_{n=0}^{\infty} e^{\alpha^2 n} z^{-n} = \frac{1}{1 - e^{\alpha^2} z^{-1}} \tag{11}$$

where $X(z)$ converges provided that $|e^{\alpha^2} z^{-1}| < 1$ which is equivalent to $|z| > |e^{\alpha^2}| = e^{\Re^2(\alpha) - \Im^2(\alpha)}$ for the ROC.

ii) For $z = e^{\jmath\omega}$, we have $z^* = z^{-1}$ and so we have

$$|G_2(z)| = |1 - (\alpha^*)^{-1} z^{-1}|$$
$$|G_2(z)| = |G_2(z)^*| = |1 - \alpha^{-1} z^{-1}| = |\alpha^{-1} z||\alpha z^{-1} - 1| = |\alpha|^{-1}||G_1(z)||$$

Thus the ratio $\frac{|G_1(z)|}{|G_2(z)|} = |\alpha|$ is independent of ω and $p = |\alpha|$. Note that the conjugation does not affect the modulus of a complex number.

iii) To find the roots of $H_1(z)$, we can either treat it as a polynomial in z^{-1} or else treat $z^2 H(z)$ as a polynomial in z. This leads to $z^{-1} = \{6, -\frac{2}{3}\}$ or else $z = \{\frac{1}{6}, -\frac{3}{2}\}$. As the zeros of $H_2(z)$ lie inside the unit circle, then we have that,

$$H_2(z) = \alpha(1 - \frac{1}{6} z^{-1})(1 + \frac{2}{3} z^{-1}).$$

In order to determine α, we substitute a suitable value of z that lies on the unit circle, e.g. $z = 1$, to obtain $|H_2(z)| = |\alpha(1 - \frac{1}{6})(1 + \frac{2}{3})| = |\alpha|\frac{25}{18} = |H_1(1)| = 25$. As a consequence, we get $\alpha = \pm 18$ and thereby

$$H_2(z) = \pm 18(1 - \frac{1}{6} z^{-1})(1 + \frac{2}{3} z^{-1}).$$

iv) Suppose that $x[n]$ is any bounded sequence with $|xn| < B < \infty\ \forall n$. We need to show that the output, $y[n] = \sum_{\tau=-\infty}^{\infty} h[\tau]x[n-\tau]$ is also bounded. Thus, we have

$$|y[n]| = |\sum_{\tau=-\infty}^{\infty} h[\tau]x[n-\tau]| \leq \sum_{\tau=-\infty}^{\infty} |h[\tau]||x[n-\tau]| < B \sum_{\tau=-\infty}^{\infty} |h[\tau]| = BS$$

We also have $H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$. If $|z| = 1$ we can write

$$|H(z)| = |\sum_{n=-\infty}^{\infty} h[n]z^{-n}| \leq \sum_{n=-\infty}^{\infty} |h[n]z^{-n}| = \sum_{n=-\infty}^{\infty} |h[n]| = S \tag{12}$$

Hence $H(z)$ is convergent for $|z| = 1$.

c) A filter, with input $x[n]$ and output $y[n]$, is defined by the difference equation

$$y[n] = \frac{2-\alpha}{1+\alpha}y[n-1] + \frac{2\alpha-1}{1+\alpha}x[n]$$

where $0 < \alpha < 1$ is a real constant.

i) Determine the system function of the filter, $H(z)$, and the impulse response, $h[n]$, for $n = -1, 0, 1$. **[3 Marks]**

ii) State the values of z at which $H(z)$ has a pole or zero. **[2 Marks]**

iii) Determine the frequency at which the filter has a gain of -3dB. **[2 Marks]**

iv) If the sample frequency is f_s, show that, for $n \geq 0$, the impulse response, $h[n]$, is equal to a sampled version $f(t) = Ae^{-t/\beta}$ and determine the values of the constants A and β. **[3 Marks]**

i) taking z-transformL $Y(z) = \frac{2-\alpha}{1+\alpha}Y(z)z^{-1} + \frac{2\alpha-1}{1+\alpha}X(z)$ from which $H(z) = \frac{Y(z)}{X(z)} = \frac{1-\frac{2-\alpha}{1+\alpha}}{1-\frac{2-\alpha}{1+\alpha}z^{-1}} = \frac{2\alpha-1}{1+\alpha-(2-\alpha)z^{-1}}$. Hence, $h[-1] = 0, h[0] = \frac{2\alpha-1}{1+\alpha}, h[1] = \frac{(2\alpha-1)(2-\alpha)}{(1+\alpha)^2}$.

ii) The system function is $H(z) = \frac{(2\alpha-1)z}{(1+\alpha)z-(2-\alpha)}$. It has a zero at $z = 0$ and a pole at $z = \frac{2-\alpha}{1+\alpha}$.

iii) We want $|H(e^{\jmath\omega})|^2 = H(e^{\jmath\omega}) \times H^*(e^{\jmath\omega}) = 0.5$. Thus

$$0.5 = \frac{1-\frac{2-\alpha}{1+\alpha}}{1-\frac{2-\alpha}{1+\alpha}e^{\jmath\omega}} \times \frac{1-\frac{2-\alpha}{1+\alpha}}{1-\frac{2-\alpha}{1+\alpha}e^{-\jmath\omega}} = \frac{(1-\frac{2-\alpha}{1+\alpha})^2}{1-2\frac{2-\alpha}{1+\alpha}\cos\omega + (\frac{2-\alpha}{1+\alpha})^2}.$$

From this, $\cos\omega = \frac{1+(\frac{2-\alpha}{1+\alpha})^2-2(1-\frac{2-\alpha}{1+\alpha})^2}{2\frac{2-\alpha}{1+\alpha}} = \frac{4\frac{2-\alpha}{1+\alpha}-1-(\frac{2-\alpha}{1+\alpha})^2}{2\frac{2-\alpha}{1+\alpha}} = 1-\frac{(1-\frac{2-\alpha}{1+\alpha})^2}{2\frac{2-\alpha}{1+\alpha}}$ and so $\omega_{3dB} = \cos^{-1}(\frac{4\frac{2-\alpha}{1+\alpha}-1-(\frac{2-\alpha}{1+\alpha})^2}{2\frac{2-\alpha}{1+\alpha}}) = \cos^{-1}(1-\frac{(1-\frac{2-\alpha}{1+\alpha})^2}{2\frac{2-\alpha}{1+\alpha}}) = \cos^{-1}(1-\frac{(2\alpha-1)^2}{2(2-\alpha)(1+\alpha)})$.

iv) For $n \geq 0$, we have from part i) that $h[n] = (1-\frac{2-\alpha}{1+\alpha})(\frac{2-\alpha}{1+\alpha})^n$. This can also be proved by induction from the given recurrence relation assuming that $x[n]$ is an impulse at $n = 0$. If we now substitute $t = \frac{n}{f_s}$ into $f(t) = Ae^{-t/\beta}$ we obtain $h[n] = f(\frac{n}{f_s}) = Ae^{-\frac{n}{\beta f_s}}$. Rearranging the later equation gives $\beta = \frac{-1}{f_s(\ln(2-\alpha)-\ln(1+\alpha))}$.

Question 3 [30/30 Marks]

a) If $P(z^2) = \frac{\alpha+z^{-2}}{1+\alpha z^{-2}}$ and $Q(z^2) = \frac{\beta+z^{-2}}{1+\beta z^{-2}}$ are stable filters. Let $H(z) = P(z^2) + z^{-1}Q(z^2)$. Determine the values of α and β so that the numerator of $H(z)$ is a scalar multiple of $(1+z^{-1})^5$. **[5 Marks]**

$$H(z) = P(z^2) + z^{-1}Q(z^2) = \frac{(\alpha+z^{-2})(1+\beta z^{-2}) + z^{-1}(\beta+z^{-2})(1+\alpha z^{-2})}{(1+\alpha z^{-2})(1+\beta z^{-2})}$$
$$= \frac{\alpha+\beta z^{-1}+(\alpha\beta+1)z^{-2}+(\alpha\beta+1)z^{-3}+\beta z^{-4}+\alpha z^{-5}}{(1+\alpha z^{-2})(1+\beta z^{-2})}$$

We need the numerator to equal $k(1+z^{-1})^5 = k(1+5z^{-1}+10z^{-2}+10z^{-3}+5z^{-4}+z^{-5})$. Identifying coefficients gives, $\alpha = k, \beta = 5k, \alpha\beta+1 = 10k$ from which $5k^2-10k+1=0$ which has the solutions $k = \frac{10\pm\sqrt{80}}{10} = 1\pm\sqrt{0.8} = \{1.8944, 0.1056\}$. Since $P(z^2)$ has a pole at $z=\sqrt{-\alpha}$, we require that $|\alpha| = |k| < 1$ for stability. Hence, $\alpha = k = 1-\sqrt{0.8} = 0.1056$ and $\beta = 5k = 5-\sqrt{20} = 0.5279$.

b) The impulse response of an antisymmetric FIR filter, $H(z)$, of order M satisfies the relation $h[n] = h[M-n]$.

i) Show that the magnitude response $|H(e^{\jmath\omega})|$ can be expressed as the absolute value of the sum of N cosine functions where $N = M/2$ if M is even and $N = (M+1)/2$ if M is odd. **[3 Marks]**

If M is odd, there is an even number of coefficients, so we can write

$$H(z) = \sum_{n=0}^{\frac{M-1}{2}} h[n]z^{-n} + \sum_{n=\frac{M+1}{2}}^{M} h[n]z^{-n} = \sum_{n=0}^{\frac{M-1}{2}} h[n]z^{-n} + \sum_{n=0}^{\frac{M-1}{2}} h[M-n]z^{-(M-n)}$$
$$= \sum_{n=0}^{\frac{M-1}{2}} h[n](z^{-n}+z^{n-M}) = z^{-0.5M}\sum_{n=0}^{\frac{M-1}{2}} h[n](z^{-n+0.5M}+z^{n-0.5M})$$

Hence, we have that

$$H(e^{\jmath\omega}) = e^{-\jmath 0.5M\omega}\sum_{n=0}^{\frac{M-1}{2}} h[n]2\cos((n-0.5M)\omega)$$

which results in

$$|H(e^{\jmath\omega})| = |\sum_{n=0}^{\frac{M-1}{2}} h[n]2\cos((n-0.5M)\omega)|.$$

The right side is the absolute value of the sum of $\frac{M+1}{2}$ cosine waves as required.

ii) Show that $H(e^{\jmath\omega})$ is necessarily zero at $\omega = \pi$ if M is odd. provide an example of a filter for which this is the case. **[2 Marks]**

When $\omega = \pi$, then $\cos((k+0.5)\omega) = 0$ for any integer k. Hence, all the summation terms are zero. An example, for $M=1$, is $H(z) = 1+z^{-1}$. For this case, $H(e^{\jmath\omega}) = 1+e^{-\jmath\omega} = e^{-\jmath 0.5\omega}2\cos(0.5\omega)$. When $\omega=\pi$, $H(e^{\jmath\omega}) = 0$.

iii) Derive an expression for the phase response, $\angle H(e^{\jmath\omega})$, and determine the group delay, $\tau_H(e^{\jmath\omega}) = \frac{-d\angle H(e^{\jmath\omega})}{d\omega}$. **[3 Marks]**

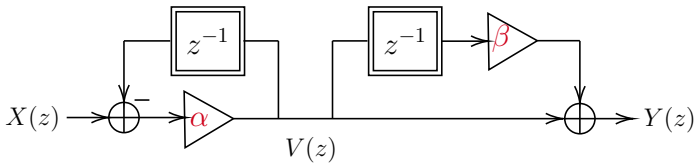
The phase response can be expressed as

$$\angle H(e^{\jmath\omega}) = \angle(e^{-\jmath 0.5M\omega}) + \frac{\pi}{2}(\text{sgn}(\sum_{n=0}^{\frac{M-1}{2}} h[n]2\cos((n-0.5M)\omega)) - 1)$$

$$= -0.5M\omega + \frac{\pi}{2}(\text{sgn}(\sum_{n=0}^{\frac{M-1}{2}} h[n]2\cos((n-0.5M)\omega)) - 1)$$

Differentiating this gives $\tau_H(e^{\jmath\omega}) = \frac{-d\angle H(e^{\jmath\omega})}{d\omega} = 0.5M$. Note that the sgn() function is piecewise constant and so its derivative is zero.

c) Let $H(z) = \frac{Y(z)}{X(z)}$ be represented by the diagram below.



i) Derive the expression of $V(z)$ **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[3 Marks]**

iii) Give conditions for which $H(z)$ realizes a first order all-pass transfer. **[2 Marks]**

iv) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[1 Mark]**

i) $V(z) = \alpha X(z) - \alpha z^{-1}V(z) \Rightarrow V(z) = \frac{\alpha X(z)}{1+\alpha z^{-1}}$.

ii) $Y(z) = V(z) + \beta z^{-1}V(z) = \frac{\alpha+\alpha\beta z^{-1}}{1+\alpha z^{-1}}X(z)$.

iii) The condition for a first order all-pass transfer is that the denominator $a[n]$ and numerator $b[n]$ of the transfer function have the same order M, and that $b[n] = a[M-n]$, which is satisfied for $H(z)$ for $\alpha \in \mathbb{R}$ and $\beta = \frac{1}{\alpha}$.

iv) The pole $-\alpha$ should be inside the unit circle, which is satisfied if $|\alpha| < 1$.

d) Consider the system with the following impulse response

$$h[n] = \left(\frac{1}{\lambda}\right)^{n+2} u[n-K],$$

where $u[n]$ denotes the step function and $\lambda > 1$.

i) Derive the phase of the system. **[3 Marks]**

ii) Derive the expression of the group delay $\tau_H(e^{\jmath\omega})$. **[3 Marks]**

iii) Find λ, K such that $\tau_H(e^{\jmath\pi}) = 3$ and $\tau_H(e^{\jmath 0}) = 1$. **[2 Marks]**

i) The group delay satisfies $\tau_H\left(e^{\jmath\omega}\right) = -\frac{d}{d\omega}\angle H\left(e^{\jmath\omega}\right)$. We proceed by computing the frequency response function $H\left(e^{\jmath\omega}\right)$ as

$$H\left(e^{\jmath\omega}\right) = e^{-K\jmath\omega}\frac{1}{\lambda^2}\frac{1}{1-\frac{1}{\lambda}e^{-\jmath\omega}}.$$

Then

$$\angle H\left(e^{\jmath\omega}\right) = -K\omega - \angle\left(1-\frac{1}{\lambda}e^{-\jmath\omega}\right) = -K\omega - \tan^{-1}\left(\frac{\frac{1}{\lambda}\sin\omega}{1-\frac{1}{\lambda}\cos\omega}\right)$$

ii) The group delay thus satisfies

$$\begin{aligned}
\tau_H\left(e^{\jmath\omega}\right) &= -K - \frac{1}{1+\frac{\frac{1}{\lambda^2}\sin^2\omega}{\left(1-\frac{1}{\lambda}\cos\omega\right)^2}}\cdot\frac{\frac{1}{\lambda}\cos\omega\left(1-\frac{1}{\lambda}\cos\omega\right)-\frac{1}{\lambda}\sin\omega\cdot\frac{1}{\lambda}\sin\omega}{\left(1-\frac{1}{\lambda}\cos\omega\right)^2} \\
&= -K - \frac{\frac{1}{\lambda}\cos\omega - \frac{1}{\lambda^2}\cos^2\omega - \frac{1}{\lambda^2}\sin^2\omega}{1+\frac{1}{\lambda^2}\cos^2\omega - \frac{2}{\lambda}\cos\omega + \frac{1}{\lambda^2}\sin^2\omega} \\
&= -K - \frac{\frac{1}{\lambda}\cos\omega - \frac{1}{\lambda^2}}{1-\frac{2}{\lambda}\cos\omega+\frac{1}{\lambda^2}} = -K - \frac{\lambda\cos\omega - 1}{\lambda^2 - 2\lambda\cos\omega + 1}
\end{aligned}$$

iii) $\tau_H\left(e^{\jmath\pi}\right) = -K - \frac{-\lambda-1}{\lambda^2+2\lambda+1} = -K + \frac{1}{\lambda+1} = 3.$
$\tau_H\left(e^{\jmath 0}\right) = -K - \frac{\lambda-1}{\lambda^2-2\lambda+1} = -K - \frac{1}{\lambda-1} = 1.$
Using the two equations above we get two solutions $\lambda = -0.62, K = -0.38$ and $\lambda = 1.62, K = -2.62$.

Question 4 [30 Marks]

a) Let $y_N[k]$ be the output of a system with impulse response $h_N[k]$ satisfying

$$y_N[k] = \sum_{n=1}^{N} \frac{x[k-n]}{n(n+1)}.$$

i) Prove that the system is linear and time-invariant. **[2 Marks]**

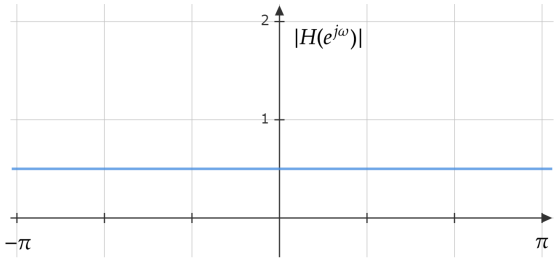
ii) Compute the system transfer function $H_N(z)$ corresponding to $h_N[k]$.
For $N = 1$, compute analytically and sketch its magnitude and phase response functions.
[3 Marks]

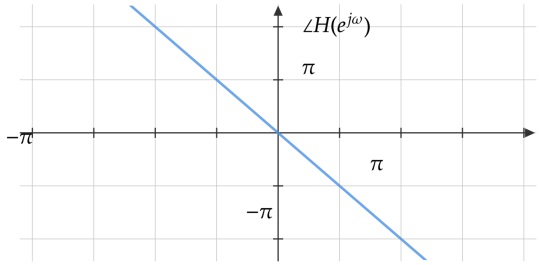
iii) Compute the step response of $H_N(z)$, denoted $y_N^0[k]$.
Find the steady state $\lim_{k\to\infty} y_N^0[k]$. **[2 Marks]**

iv) Let E_x^N be defined as $E_x^N = \sum_{k=-N}^{N} |x[k]|^2$, such that $E_x = \lim_{N\to\infty} E_x^N$ is the "energy" of a sequence $x[k]$. Calculate the expression of $E_{y_N^0}^N$ as a function of N.
Compute $E_y = \lim_{N\to\infty} E_{y_N^0}^N$. **[3 Marks]**

i) $\mathcal{H}(\alpha x_1 + x_2)[k] = \sum_{n=1}^{N} \frac{\alpha x_1[k-n]+x_2[k-n]}{n(n+1)} = \alpha \sum_{n=1}^{N} \frac{x_1[k-n]}{n(n+1)} + \sum_{n=1}^{N} \frac{x_2[k-n]}{n(n+1)}$,
$\mathcal{H}(x[\cdot - m])[k] = \sum_{n=1}^{N} \frac{x[k-n-m]}{n(n+1)} = \mathcal{H}x[k-m]$.

ii) The transfer function satisfies $H_N(z) = \sum_{n=1}^{N} \frac{z^{-n}}{n(n+1)}$. For $N = 1$ we have $H_1(z) = \frac{1}{2}z^{-1}$. Then $|H_1(e^{j\omega})| = \frac{1}{2}$ and $\angle H(e^{j\omega}) = -\omega$.
The two expressions are depicted below.





iii) Here $x[k]$ is the step function, and thus $x[k] = 0, k < 0$. It follows that

$$y_N^0[k] = 0, k < 1.$$

For $k \geq 1$ we have

$$y_N^0[k] = \sum_{n=1}^{\min\{N,k\}} \frac{1}{n(n+1)} = \sum_{n=1}^{\min\{N,k\}} \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{\min\{N,k\}+1}.$$

Lastly,

$$\lim_{k\to\infty} y_N^0[k] = 1 - \frac{1}{N+1}$$

.

iv) $E_{y_N^0}^N$ has the expression

$$E_{y_N^0}^N = \sum_{k=-N}^{N} \left| 1 - \frac{1}{m_{N,k}+1} \right|^2,$$

where $m_{N,k} \triangleq \min\{N, k\}$, and thus

$$E_{y_N^0}^N = \sum_{k=-N}^{N} \left| 1 - \frac{1}{k+1} \right|^2.$$

Given that $\left|1 - \frac{1}{k+1}\right|^2$ does not converge to 0, then the sum diverges and thus $E_y = \infty$.

b) Let $H(z) = H_1(z) + z^{-1}H_2(z)$ where $H_1(z)$ and $H_2(z)$ are two filters satisfying

$$H_1(z) = \beta/2 + z^{-1} - z^{-2} - z^{-4}, \tag{13}$$
$$H_2(z) = 2 + (\gamma - 3)z^{-1} - 4z^{-3} + 3z^{-4} - 2z^{-5}. \tag{14}$$

i) Compute β and γ such that $h[n]$ has a linear phase response. **[4 Marks]**

ii) Derive analytically the phase and gain response of $h[n]$. **[4 Marks]**

i) We have that

$$H(z) = \beta/2 + 3z^{-1} + (\gamma - 4)z^{-2} - 5z^{-4} + 3z^{-5} - 2z^{-6}.$$

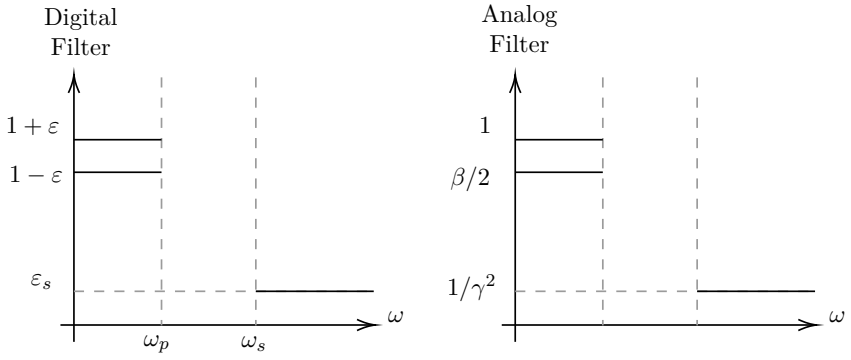
We need $h[n]$ to be anti-symmetric or symmetric. Given that $h[1] = h[5]$, the filter can only be symmetric, which is satisfied if $\beta = -4$ and $\gamma = -1$.

ii) We have that

$$\begin{aligned} H(e^{\jmath\omega}) &= -2 + 3e^{-\jmath\omega} - 5e^{-2\jmath\omega} - 5e^{-4\jmath\omega} + 3e^{-5\jmath\omega} - 2e^{-6\jmath\omega} \\ &= e^{-3\jmath\omega}\left[-2\left(e^{3\jmath\omega} + e^{-3\jmath\omega}\right) + 3\left(e^{2\jmath\omega} + e^{-2\jmath\omega}\right) - 5\left(e^{\jmath\omega} + e^{-\jmath\omega}\right)\right] \\ &= e^{-3\jmath\omega}\left[-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega)\right]. \end{aligned}$$

It follows that $|H(e^{\jmath\omega})| = |-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega)|$ and $\angle H(e^{\jmath\omega}) = -3\omega + (1 - \text{sign}(-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega))) \cdot \frac{\pi}{2}$.

c) Filter specifications for low-pass digital and analog filters are shown below, respectively.



Using the Bilinear Transformation, the goal is to design a discrete-time IIR filter that satisfies the frequency response constraints. Furthermore, it is desirable that the peak gain (digital filter) in the passband is unity, that is,

$$\max_{\omega\in[0,\omega_p]} |H\left(e^{\jmath\omega}\right)| = 1.$$

Given that for a digital filter, the filter order is estimated by,

$$M \approx -\frac{1}{|\omega_s - \omega_p|}\left(\frac{56}{10} + \frac{43}{10}\log_{10}\left(\varepsilon_s\varepsilon\right)\right),$$

compute the filter order in terms of analog filter parameters μ and λ. **[4 Marks]**

Clearly,

$$1-\varepsilon \leqslant |H\left(e^{\jmath\omega}\right)| \leqslant 1+\varepsilon \Rightarrow \frac{1-\varepsilon}{1+\varepsilon} \leqslant \frac{|H\left(e^{\jmath\omega}\right)|}{1+\varepsilon} \leqslant 1$$

which also satisfies the condition that, $\max_{\omega\in[0,\omega_p]} |H\left(e^{\jmath\omega}\right)| = 1$. Matching the analog filter specification, we obtain,

$$\frac{1-\varepsilon}{1+\varepsilon} = \frac{\beta}{2} \Rightarrow \varepsilon = \frac{2-\beta}{2+\beta}.$$

Also since the peak passband gain is unity, we have,

$$\frac{1}{\gamma^2} = \frac{\varepsilon_s}{1+\varepsilon} \Rightarrow \varepsilon_s = \frac{1+\varepsilon}{\gamma^2} = \frac{4}{2+\beta}\frac{1}{\gamma^2}.$$

Hence the filter order is estimated to be,

$$M \approx -\frac{1}{|\omega_s - \omega_p|}\left(\frac{56}{10} + \frac{43}{10}\log_{10}\left(\frac{4(2-\beta)}{(2+\beta)^2\gamma^2}\right)\right).$$

d) Let $H\left(\jmath\omega\right)$ be an all-pass filter with a non-linear phase response $\angle H\left(\jmath\omega\right) = \beta\omega^3 - \frac{\omega^2}{2}$. Let $H_1(j\omega)$ be a filter with all-pass gain and phase given by

$$\angle H_1\left(\jmath\omega\right) = \angle H\left(\jmath\omega_0\right) + \frac{\left(\omega-\omega_0\right)^2}{2}\frac{d^2\angle H}{d\omega^2}\left(\jmath\omega_0\right). \tag{15}$$

i) Compute the frequency response $H_1\left(\jmath\omega\right)$. **[2 Marks]**

ii) Let $H_2\left(\jmath\omega\right)$ be an all-pass filter with phase

$$\angle H_2\left(\jmath\omega\right) = \angle H\left(\jmath\omega_0\right) + \frac{\left(\omega-\omega_0\right)^2}{2}\frac{d^2\angle H}{d\omega^2}\left(\jmath\omega_0\right) + \frac{\left(\omega-\omega_0\right)^3}{6}\frac{d^3\angle H}{d\omega^3}\left(\jmath\omega_0\right). \tag{16}$$

For $\omega_0 = 0$, compute $H_2\left(\jmath\omega\right) - H\left(\jmath\omega\right)$. **[3 Marks]**

iii) Find β as a function of ω_0 such that $\omega_0 \cdot \tau_H\left(\jmath\omega_0\right) = \angle H_1\left(\jmath\omega_0\right)$ **[3 Marks]**

i) We have that $|H_1\left(\jmath\omega\right)| = 1$ and

$$\angle H_1\left(\jmath\omega\right) = \beta\omega_0^3 - \frac{\omega_0^2}{2} + \frac{\left(\omega-\omega_0\right)^2}{2}\left(6\beta\omega_0 - 1\right).$$

Thus $H_1\left(\jmath\omega\right) = e^{\jmath\angle H_1(\jmath\omega)}$.

ii) The third term in (16) satisfies

$$\frac{(\omega-\omega_0)^3}{6}\frac{d^3\angle H}{d\omega^3}(\jmath\omega_0)=\beta(\omega-\omega_0)^3,$$

and thus

$$H_2(\jmath\omega)=H_1(\jmath\omega)\cdot e^{\jmath\beta(\omega-\omega_0)^3}.$$

For $\omega_0=0$, we get

$$H_2(\jmath\omega)=e^{\jmath\left(\beta\omega^3-\omega^2/2\right)},$$

and thus

$$H_2(\jmath\omega)-H(\jmath\omega)=0.$$

iii) The group delay satisfies

$$\tau_H(\jmath\omega_0)=-\frac{d\angle H(\jmath\omega_0)}{d\omega}=-3\beta\omega_0^2+\omega_0.$$

It follows that the following are equivalent

$$\begin{aligned}\omega_0\cdot\tau_H(\jmath\omega_0)&=\angle H_1(\jmath\omega_0)\\ \omega_0^2\left[-3\beta\omega_0+1\right]&=\omega_0^2\left[\beta\omega_0-1/2\right]\\ \beta&=\frac{3}{8\omega_0}.\end{aligned}$$